Did Quiroga’s method ever work for us?

Answer: the printed adaptive rule is not reproduced; modified fixed-penalty FGD does recover accurately.

“Never worked” is too broad. Our implementation of the printed adaFGD step has not passed the full recovery gate in the audited campaigns. That is different from saying that fixed-penalty factorized gradient descent fails, or that the paper is wrong.

Method / protocol	Qubits	Observed result
Printed-rule adaFGD: latest screen	2	0/44 pass at 10k updates; 14 finite misses, 30 numerical failures
Printed-rule adaFGD: earlier long tests	2	0/4 pass at 100k updates; fidelity 99.53–99.87%
Our exact factor-ray line-search FGD	2	16/16 pass with one frozen fixed-penalty setting
Earlier Armijo-backtracking FGD	4	4/4 pass; different historical initialization and tuning
Latest independently tuned FRAMES	4	0/2 joint passes; fidelity and TP pass, RMSE narrowly misses

The successful FGD variants retain a fixed TP penalty but replace step selection. Exact-ray FGD minimizes the objective along the factor-gradient direction; Armijo FGD uses sufficient-decrease backtracking. Neither should be labelled “published Quiroga adaFGD”.

Our joint gate requires fidelity ≥ 0.99, full-design RMSE $\leq 10^{-6}$ and normalized TP residual $\leq 10^{-3}$. The long printed-rule runs exceeded 99.5% fidelity but still failed both data fit and TP. Thus “no recovery at all” is also misleading.

All results above are noiseless rank-one tests. Different dimensions, budgets and protocols prevent a direct speed or superiority ranking from this table.

Meeting update: what changed over the last week

Evidence through 12 October, checked 14 October 2026. No new experiments launched for this briefing.

8 October — a working modified fixed-penalty baseline

Exact-ray FGD passed 2 calibration + 6 confirmation + 8 historical revalidation channels at n=2. Worst RMSE: 1.61e-10; worst normalized TP: 7.80e-10; fidelity is essentially one. The eight historical channels are not an unseen-test set. Printed-rule reproduction remained unresolved.

12 October — substantial FRAMES improvement at n=4

Completed a 12-setting penalty/step/sampling screen, long-budget checks and a separately declared two-phase step test, all locally. Latest paired RMSE: 1.688e−6 and 1.795e−6; fidelity >0.99999999; normalized TP: 5.04e−5 and 5.37e−5. These miss only the 1e−6 RMSE gate. Confirmation was skipped; the proposed longer precision run has not been launched.

What we can safely say to the coauthors

“We can solve the corrected physical sensing problem accurately with fixed-penalty FGD when we modify the step selection. We have not yet reproduced the published adaptive-rule results under our full recovery criterion. FRAMES tuning has substantially improved four-qubit recovery, but the strict gate and the final best-for-best comparison remain unfinished.”

Open items and cautions

Obtain an author reference implementation/configuration to resolve the printed-rule discrepancy; do not interpret our negative runs as disproving the paper. Keep historical n=4 Armijo evidence separate from the current protocol. Legacy high-qubit local-Pauli results remain a separate benchmark. A fixed-setting longer FRAMES precision test is the next proposed local experiment.

Evidence: Oct8 audit_screen/confirmation/revalidation and coauthor briefing; Oct7 Gell-Mann audit; Oct12 FRAMES manifests and calibration audit. Archived sources and hashes are listed in the accompanying evidence JSON.